

Docker



helle

Overview

Topic	Sub-Topics	Duration (in Days)	Duration (in hours)	
Networking	Advanced Networking Concepts, Custom Networks, Overlay Networks, Network Namespaces, Docker Swarm Networking, Service Discovery	0.5	4	Total duration: 2 weeks
Docker Compose	Advanced Compose Configuration, Managing Complex Applications, Deploying with Compose, Environment Variables, Compose and Swarm	0.5	4	
Registry	Setting Up Private Registries, Registry Storage Backends, Registry Authentication, Image Tagging, Promoting Images Across Environments	0.5	4	
Docker CLI	Advanced Docker CLI Usage, Debugging with Docker CLI, Performance Monitoring, Docker CLI Plugins	0.5	4	
Volumes	Persistent Storage, Named Volumes, Bind Mounts, Volume Drivers, Backup and Restore, Data Management in Docker	0.5	4	
Security	Image Security, Container Security, Docker Content Trust, Seccomp, SELinux, User Namespaces, Secrets Management	0.5	4	
Multi-stage builds	Optimizing Dockerfiles, Reducing Image Size, Build Caching, Multi-Stage Builds for CI/CD Pipelines	0.5	4	
Best Practices	Dockerfile Best Practices, Managing Resources, Efficient Caching, Minimizing Layers, Keeping Containers Stateless	0.5	4	
Assignment + Level Up		6	48	

Introduction

Deepen your Docker expertise by creating multi-stage builds, setting up networks, implementing best practices and managing volumes.

Prerequisites

Docker 20.10 or later, Docker compose 1.29 or later, Basic understanding of Linux command-line operations, Basic knowledge of web application development.

Usage

Create multi-stage Dockerfiles, setting up networks, implementing best practices and managing volumes.

Learning Expectations

Gain practical experience in Docker, understanding Docker Compose and networking, and skills in using Docker registries.

Library Version

Docker 20.10+, Docker Compose 1.29+

Course Material

Topic	Documentation
Networking	https://docs.docker.com/network/
Docker Compose	https://docs.docker.com/compose/gettingstarted/
Registry	https://docs.docker.com/guides/docker-concepts/the-basics/what-is-a-registry/
Docker CLI	https://docs.docker.com/engine/reference/commandline/cli/
Volumes	https://docs.docker.com/storage/volumes/
Security	https://cheatsheetseries.owasp.org/cheatsheets/Docker_Security_Cheat_Sheet.html
Multi-stage builds	https://docs.docker.com/build/building/multi-stage/
Best practices	https://docs.docker.com/build/building/best-practices/

Learning Modules

Additional Resources

- <https://www.youtube.com/watch?v=iqqDU2crIEQ>
- <https://www.youtube.com/watch?v=SXwC9fSwct8>
- <https://www.youtube.com/watch?v=pg19Z8LL06w>
- <https://www.youtube.com/watch?v=pTFZFxd4hOI&t=9s>
- <https://www.youtube.com/watch?v=8vXoMqWgbQQ>
- <https://www.youtube.com/watch?v=VOK06Q4QqvE>
- <https://www.youtube.com/watch?v=9xz9sBjZwXs>
- <https://www.youtube.com/watch?v=OVYNUg0aODw>
- [Docker Tutorial for Beginners: Basics, Architecture, Containers](#)
- [Docker Tutorial-The Complete Beginner-to-Advanced Guide | Medium](#)

- [Docker - Quick Guide](#)
- [Docker Tutorials | Advanced Docker Concepts - StackBay](#)

Level Up

Docker

<https://nagarro.sharepoint.com/sites/university/SitePages/Pathways.aspx?pathways=Dockers%20fundamentals>

Additional Reference Links:

LinkedIn Learning

- <https://www.linkedin.com/learning/docker-for-developers-14493163/docker-for-developers>
- <https://www.linkedin.com/learning/learning-docker-17236240/get-up-and-running-quick-with-docker>
- <https://www.linkedin.com/learning/docker-essential-training/diving-deeper-into-docker>
- <https://www.linkedin.com/learning/paths/docker-foundations-professional-certificate>
- <https://www.linkedin.com/learning/introduction-to-docker-for-java-developers/layering-in-docker-images>